DevOps Engineer Challenge - Project Report

End-to-End DevOps: Deploying a React Todo App on AWS with Nginx, PM2, Jenkins, Docker, and CI/CD

1. INTRODUCTION :

This document provides a detailed walkthrough of the **DevOps Engineer Challenge**, covering the setup, execution, and deployment of a React-based Todo application on an AWS EC2 instance. The challenge involved user management, application deployment using Nginx and PM2, CI/CD pipeline integration with Jenkins, security enhancements, and Dockerization of the application.

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2. OBJECTIVES :

The primary objectives of this challenge were:

- Setting up an AWS EC2 instance and configuring a secure environment.
- Deploying a React Todo application with Nginx and PM2.
- Configuring a CI/CD pipeline using Jenkins for automated deployment.
- Enhancing security using UFW (Uncomplicated Firewall).
- Dockerizing the application and managing it using Docker Compose.

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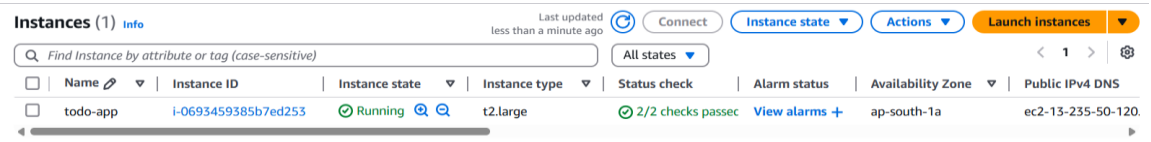
3. ENVIRONMENT SETUP :

- Operating System: Ubuntu Linux
- Tools & Technologies: Node.js, npm, PM2, Nginx, Jenkins, AWS S3, Docker, Docker Compose
- Cloud Provider: AWS (EC2 instance)

3.1 AWS EC2 Instance Configuration

1. Launched an **Ubuntu-based EC2 instance** in AWS.



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2. Connected to the instance using SSH:

   ssh -i your-key.pem ubuntu@<EC2-Public-IP>

3. Updated the system:

   sudo apt update && apt upgrade -y

3.2 User Management

1. Created a new user named **DevOps**:

   sudo adduser -m DevOps

2. Granted sudo privileges to the user:

   sudo usermod -aG sudo DevOps

3. Remove Password authentication:

   sudo passwd -d DevOps

4. Switch to DevOps User:

   sudo su - DevOps

```
DevOps@ip-172-31-38-158:~$ whoami
DevOps
DevOps@ip-172-31-38-158:~$ groups DevOps
DevOps : DevOps sudo
DevOps@ip-172-31-38-158:~$
```

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4. REACT APPLICATION DEPLOYMENT :

4.1 Install Required Software

1. Installed **Node.js** and **npm**:

   sudo apt install nodejs npm -y

2. Verified installation:

   node –version

   npm --version

4.2 Clone & Build the React Project

1. Cloned the repository:

   sudo git clone https://github.com/kabirbaidhya/react-todo- app.git /opt/checkout/react-todo-app

2. Installed project dependencies:

   cd /opt/checkout/react-todo-add

   sudo npm install

3. Built the project:

   sudo npm run build

4. Created a deployment directory and moved the build files:

   sudo mv build react

   sudo mkdir -p /opt/deployment

   sudo cp -r react /opt/deployment

4.3 Deploy Using PM2

1. Installed PM2:

   sudo npm install -g serve pm2

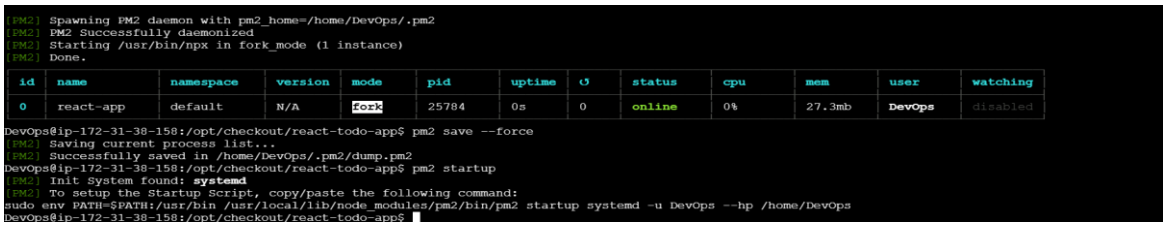
2. Started serving the React app:

   pm2 start npx --name react-app -- serve -s /opt/deployment/react

3. Configured PM2 to start on boot:

   pm2 save --force

   pm2 startup



```
[PM2] Spawning PM2 daemon with pm2_home=/home/DevOps/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /usr/bin/npx in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	react-app	default	N/A	fork	25784	0s	0	online	0%	27.3mb	DevOps	disabled

```
DevOps@ip-172-31-38-158:/opt/checkout/react-todo-app$ pm2 save --force
[PM2] Saving current process list...
[PM2] Successfully saved in /home/DevOps/.pm2/dump.pm2
DevOps@ip-172-31-38-158:/opt/checkout/react-todo-app$ pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/local/lib/node_modules/pm2/bin/pm2 startup systemd -u DevOps --hp /home/DevOps
DevOps@ip-172-31-38-158:/opt/checkout/react-todo-app$
```

5. NGINX CONFIGURATION & SECURITY :

5.1 Install & Configure Nginx

1. Installed Nginx:

   sudo apt install nginx -y

2. Created a new Nginx configuration file:

   sudo nano /etc/nginx/sites-available/default

3. Added the following content:

```
server {
    listen 80;
    location / {
        proxy_pass http://localhost:3000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }
}
```

4. Enabled the configuration and restarted Nginx:

   sudo nginx -t

   sudo systemctl restart nginx

5.2 Security Enhancements

1. Configured **UFW firewall** to allow only necessary ports:

   sudo ufw allow 80

sudo ufw allow 443

sudo ufw allow 3000

sudo ufw allow 22

sudo ufw enable

2. Verified firewall settings:

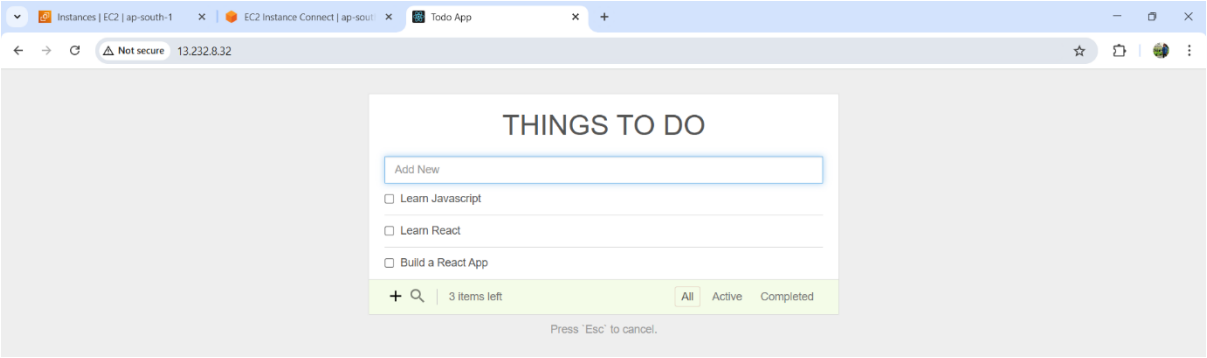
sudo ufw status

5.3: Verify the Deployment

- Open a browser and visit**:**

http://<EC2-Public-IP>

- You should see the React Todo App:



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6. CI/CD PIPELINE WITH JENKINS :

6.1 Install Jenkins

1. Installed Java (required for Jenkins):

sudo apt install fontconfig openjdk-17-jre

2. Installed Jenkins:

***sudo wget -O /usr/share/keyrings/jenkins-keyring.asc ***

https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key

***echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]" ***

***https://pkg.jenkins.io/debian-stable binary/ | sudo tee ***

/etc/apt/sources.list.d/jenkins.list > /dev/null

sudo apt-get update

sudo apt-get install jenkins

3. Add Jenkins to the sudo group:

   sudo usermod -aG sudo jenkins

4. Allow Jenkins to use sudo without password:

   sudo visudo

   Add this line:

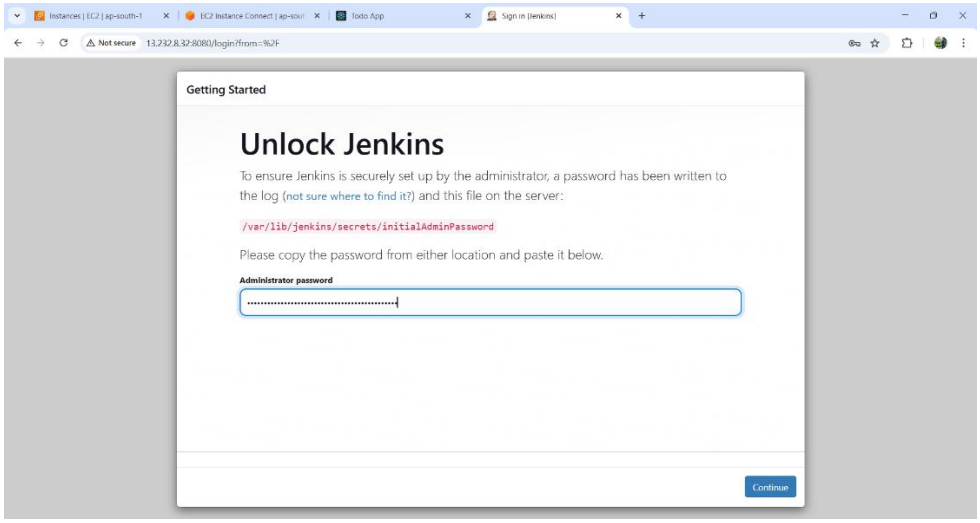
   jenkins ALL=(ALL) NOPASSWD: ALL

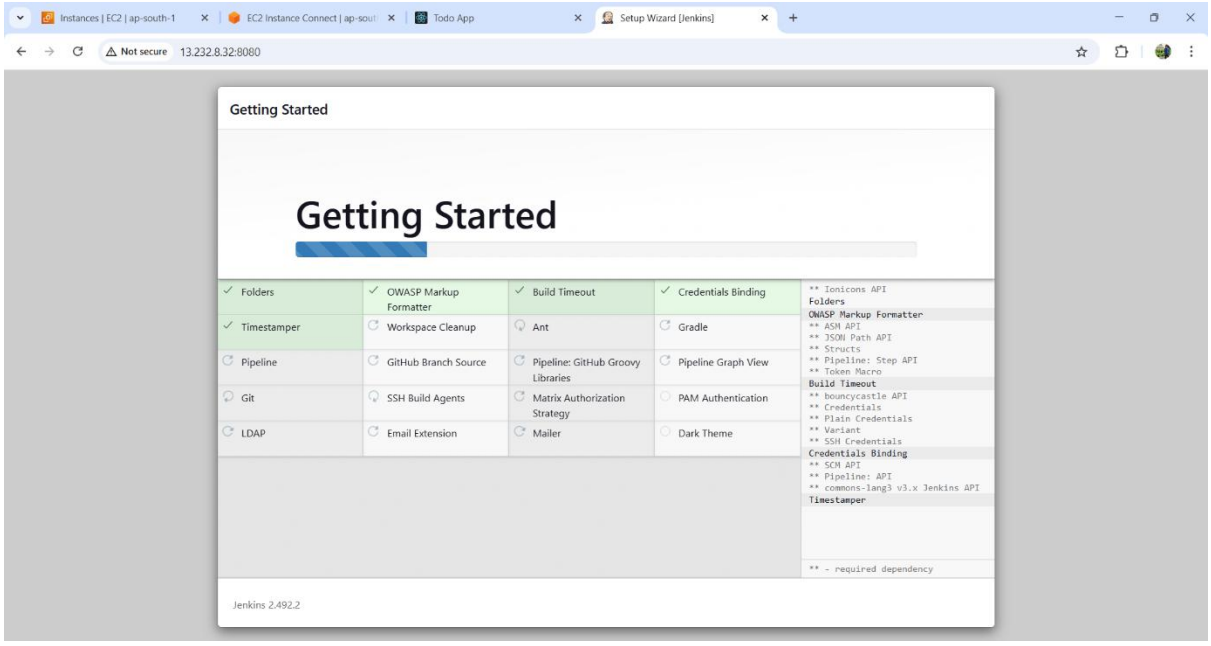
5. Open a browser and access Jenkins:

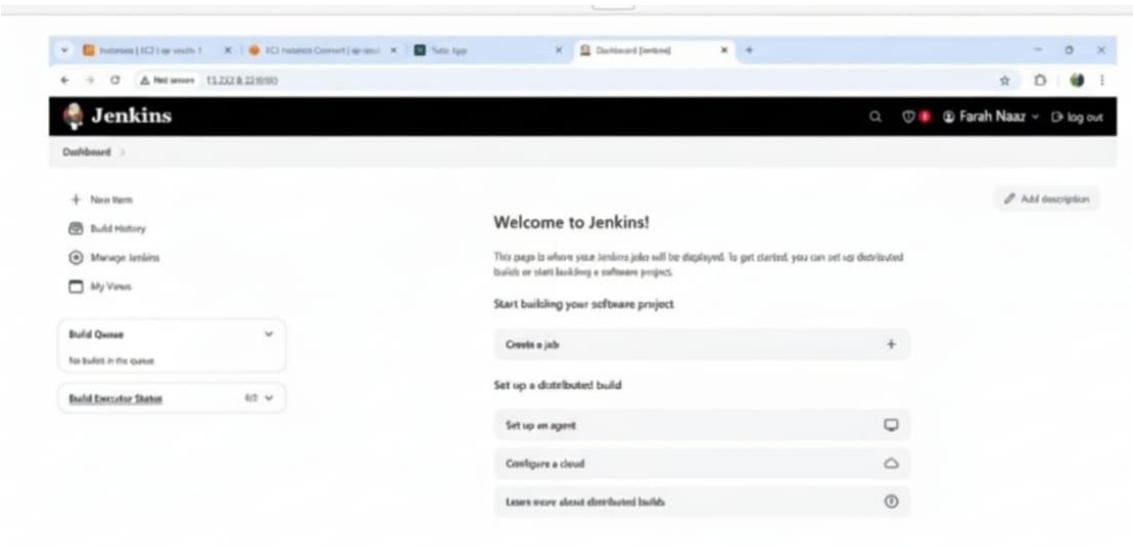
   http://<EC2-Public-IP>:8080

6. Retrieved the Jenkins admin password:

   sudo cat /var/lib/jenkins/secrets/initialAdminPassword







6.2 Install Plugins

After initial setup:

1. Go to Manage Jenkins > Plugins > Available Plugins
2. Search for and install:
    - Pipeline: AWS Steps
    - AWS Credentials Plugin

6.3 Configure AWS Credentials in Jenkins

1. Go to Manage Jenkins > Credentials > Global > Add Credentials

2. Select AWS Credentials
3. Enter:
    - ID: aws-credentials-id
    - Access Key ID: (from AWS)
    - Secret Access Key: (from AWS)

6.4 Create a Jenkins Pipeline Job

1. Go to **Jenkins Dashboard → New Item**.
2. Select **Pipeline**, name it React-Todo-CICD, and click **OK**.
3. Scroll down to the **Pipeline** section and select **Pipeline Script**.
4. Add the following pipeline script:

```
pipeline {
    agent any
    environment {
        APP_NAME = 'react-app'
        GIT_REPO = 'https://github.com/kabirbaidhya/react-todo-app.git'
        CHECKOUT_DIR = '/opt/checkout/react-todo-add'
        BUILD_DIR = '/opt/deployment/react'
        S3_BUCKET = 'react-tod-app-s3bucket-devops-challenge'
    }
    stages {
        stage('Stop Existing Deployment') {
            steps {
                sh 'sudo -u DevOps pm2 stop all || echo "App not running"'
```

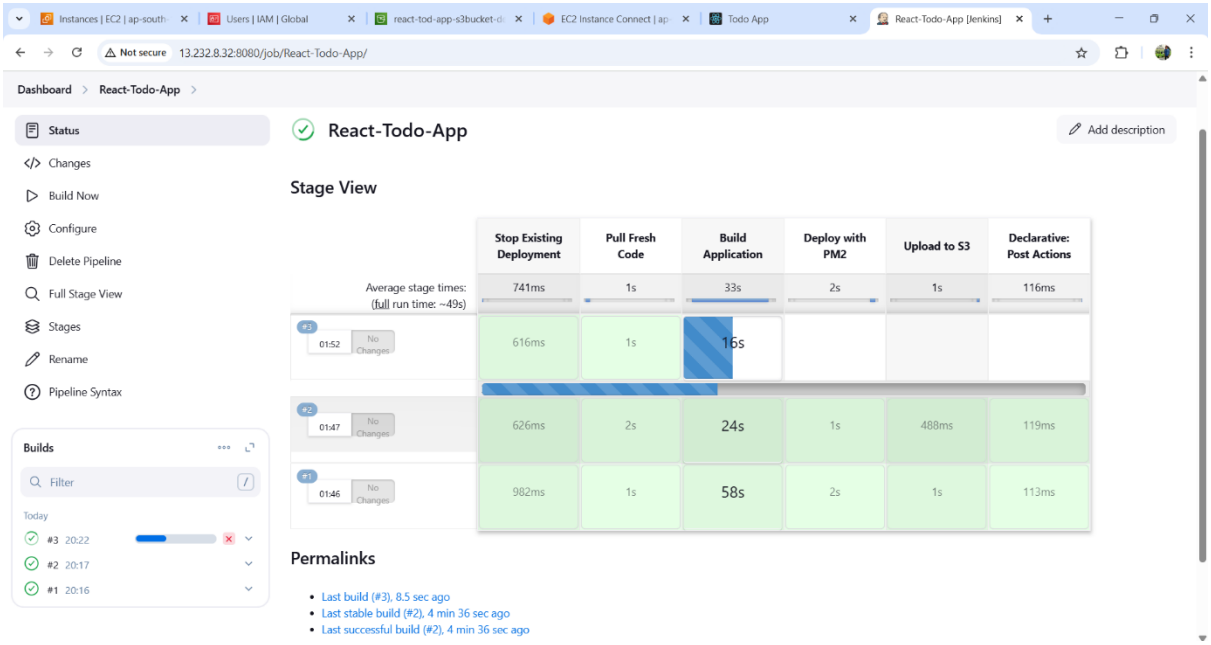
```
        }
    }
    stage('Pull Fresh Code') {
        steps {
            sh '''
            sudo chown -R jenkins:jenkins /opt/checkout/ /opt/deployment/
            sudo rm -rf $CHECKOUT_DIR || true
            sudo git clone $GIT_REPO $CHECKOUT_DIR
            '''
        }
    }
    stage('Build Application') {
        steps {
            dir("${CHECKOUT_DIR}") {
                sh 'sudo npm install'
                sh 'sudo npm run build'
            }
            sh '''
            sudo rm -rf $BUILD_DIR || true
            sudo mkdir -p $BUILD_DIR
            sudo cp -r $CHECKOUT_DIR/build/* $BUILD_DIR/
```

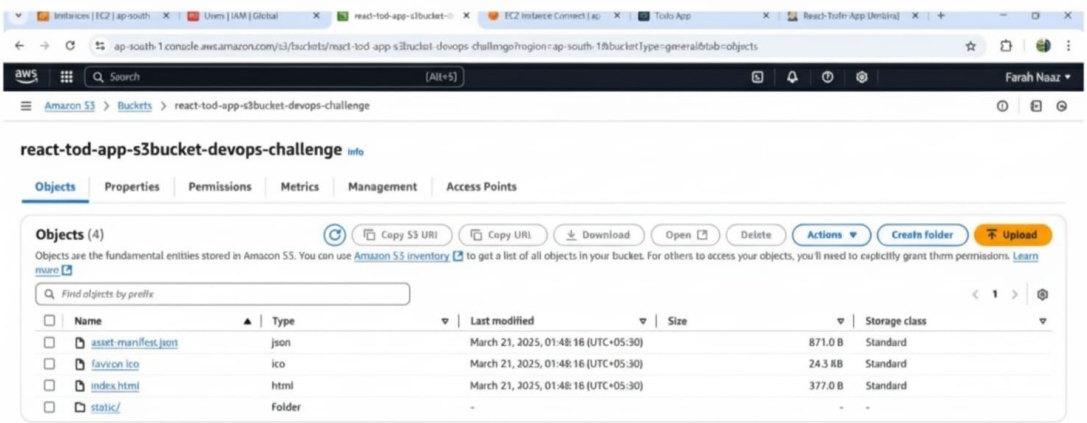
```
            '''
        }
    }
    stage('Deploy with PM2') {
        steps {
            sh '''
            sudo pm2 start npx --name $APP_NAME -- serve -
s $BUILD_DIR
            sudo pm2 save --force
            sudo pm2 startup
            '''
        }
    }
    stage('Upload to S3') {
        steps {
            withAWS(credentials: '390402563998') {
                s3Upload(
                    bucket: "$S3_BUCKET",
                    file: "$BUILD_DIR",
                )
            }
        }
    }
```

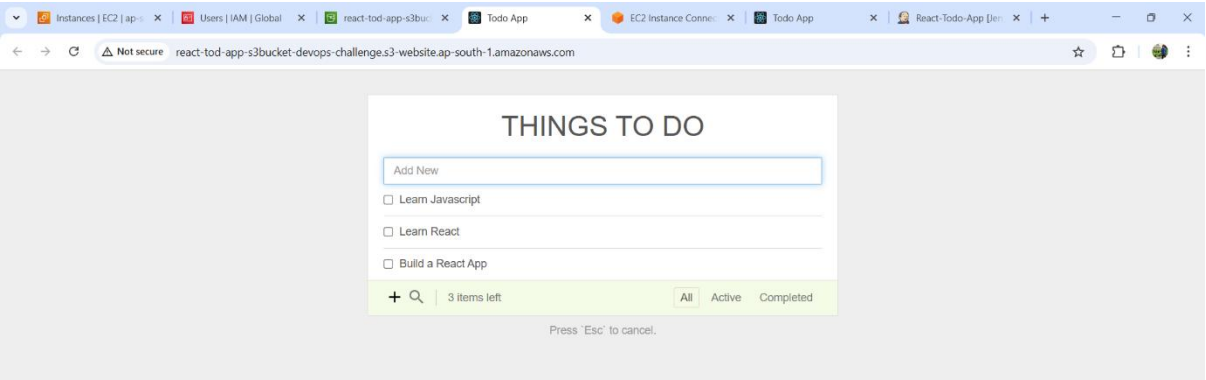
```
    }
    post {
        success { echo 'Deployment and S3 upload successful!' }
        failure { echo 'Deployment or S3 upload failed.' }
    }
}
```

6.5 Run and Monitor the Pipeline

- Click Build Now to start the pipeline.
- View logs under Build Console Output.







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7. SHELL SCRIPTING FOR JAVA INSTALLATION :

1. Created a Bash script for installing OpenJDK 1.8:

```
sudo nano /opt/scripts/install_java.sh
```

2. Added the following content:

```
#!/bin/bash
echo "$(date) - Starting Java Installation" |
tee -a    /opt/logs/script_logs.log
sudo apt update && sudo apt install -y openjdk-8-jdk
echo "$(date) - Java Installed" | tee -a /opt/logs/script_logs.log
```

3. Made the script executable and ran it:

```
chmod +x /opt/scripts/install_java.sh
sudo /opt/scripts/install_java.sh
```

---

8. DOCKERIZATION OF THE REACT APPLICATION :

8.1 Install Docker & Docker Compose

```
sudo apt install docker.io -y
sudo usermod -aG docker $USER
sudo reboot
```

8.2 Install Docker Compose:

```
mkdir -p ~/.docker/cli-plugins/
curl -SL https://github.com/docker/compose/releases/download/v2.3.3/docker-compose-linux-x86_64 -o ~/.docker/cli-plugins/docker-compose
chmod +x ~/.docker/cli-plugins/docker-compose
```

8.3 Clone the repository and configure Nginx:

```
git clone https://github.com/kabirbaidhya/react-todo-app.git
cd react-todo-app
mkdir .docker && cd .docker
```

8.4 Create and configure nginx.conf:

```
events { worker_connections 1024; }
http {
  include /etc/nginx/mime.types;
  server {
    listen 80;
    location / {
      root /usr/share/nginx/html;
      index index.html;
      try_files $uri $uri/ /index.html;
    }
    error_page 404 /index.html;
```

```
  }
}
```

8.5 Create Dockerfile:

```
nano Dockerfile
```

Add Content:

```
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build
FROM nginx:1.25.1
COPY .docker/nginx.conf /etc/nginx/nginx.conf
COPY --from=build /app/build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

8.6 Create a Docker Compose File

```
sudo nano /opt/deployment/react/docker-compose.yml
```

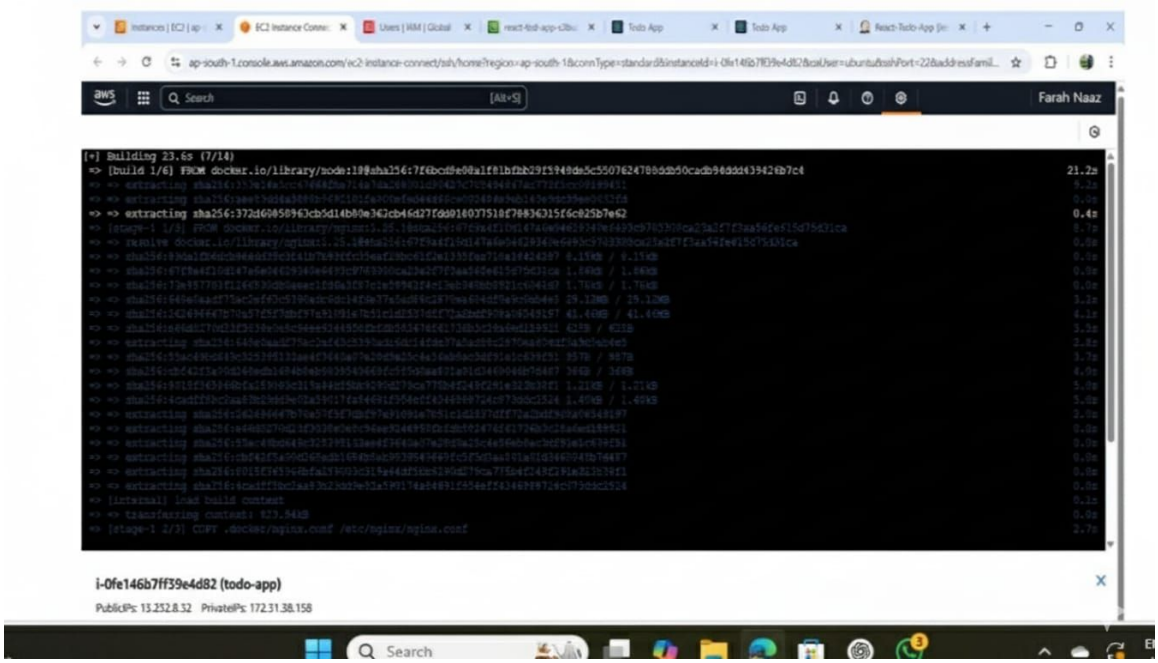
Content:

```
version: '3'
services:
  react-app:
```

```
build:
  context: .
  dockerfile: Dockerfile
ports:
  - "3000:80"
```

8.7 Run the Container

```
docker-compose up -d
```




```
docker ps
```

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9. EXECUTION & TESTING :

- Access the app via **http://<EC2-Public-IP>**.
- Verify **Dockerized version** via **http://<EC2-Public-IP>:3000**.
- Check **Jenkins pipeline execution logs**.

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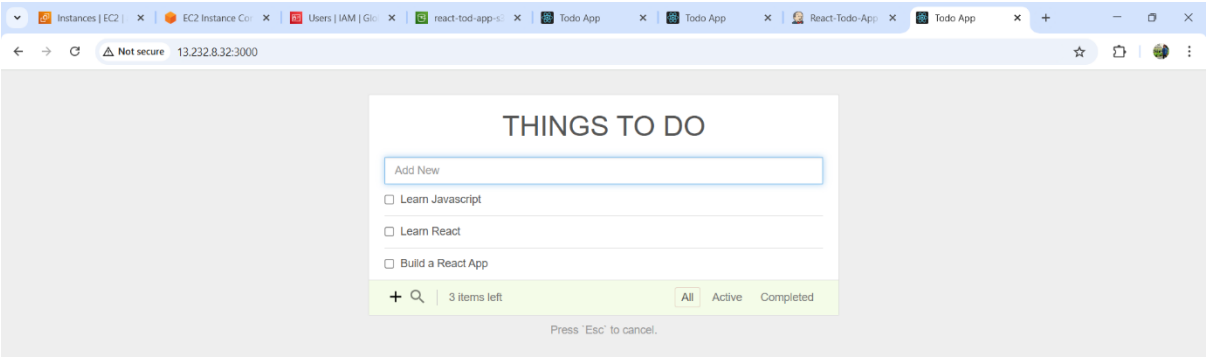
10. OUTPUT SCREENSHOT:



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11. CONCLUSION :

This challenge provided hands-on experience in DevOps workflows, cloud deployment, CI/CD automation, and containerization. The project successfully established a **secure, automated deployment pipeline** for the React application.

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12. FUTURE ENHANCEMENTS :

- Integrate Kubernetes for container orchestration.
- Implement monitoring with Prometheus and Grafana.
- Automate infrastructure provisioning using Terraform.